

山 东 滨 农 科 技 有 限 公 司
SHANDONG BINNONG TECHNOLOGY CO., LTD.

Add: NO.518, Yongxin Road, Binbei Town, Binzhou, Shandong, China

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Http:// www.binnong.co

Material Safety Data Sheet

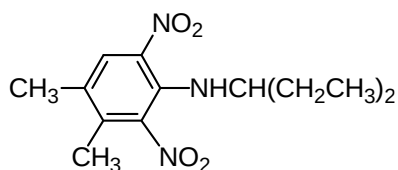
1. Chemical Product Identification

Product Name: Pendimethalin 330G/L EC

Molecular formula: $C_{13}H_{19}N_3O_4$

Molecular weight: 281.3

Structural formula:



Chemical Name: N-(1-ethylpropyl)-2,6-dinitro-3,4-xylidine

CAS No.: 40487-42-1

Chemical group: Dinitroaniline herbicides

Manufacturer: Shandong Binnong Technology Co., Ltd.

Address: No.518, Yongxin Road, Binbei, Binzhou, Shandong, China

2. Composition / Information on Ingredients

Ingredient		CAS No	Percent
Active ingredient	Pendimethalin	40487-42-1	33%
Inert ingredient	Other ingredients, including water	//	Up to 100%

3. Hazards Identification

ACUTE EFFECTS:

Swallowed: Data suggests that this product should present no significant problems. However, the solvent can be of concern if vomiting has occurred; aspiration of the vomit into the lungs can cause mild to severe pulmonary injury and possibly death.

Eye: This product is no irritating to eyes.

Skin: This product is no irritating to skin.

Inhalation: Avoid working in, or inhaling spray mist.

4. First Aid Measures

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If in eyes: Hold eye open and rinse slowly and gently with water for 15-20 minutes. Remove contact lenses, if present, after first 5 minutes, then continue rinsing eye. Call a poison control center or doctor for treatment advice.

If on skin: Take off contaminated clothing. Rinse skin immediately with plenty of water for 15-20 minutes. Call a poison control center or doctor for treatment advice.

If swallowed: Call a poison control center or doctor immediately for treatment advice. Have person sip a glass of water if able to swallow. Do not induce vomiting unless told to by a poison control center or doctor. Do not give anything to an unconscious person

If inhaled: Move person to fresh air. If person is not breathing, call 911 or an ambulance, and then give artificial respiration, preferably mouth-to-mouth if possible. Call a poison control center or doctor for further treatment advice.

Note to physician: Pendimethalin is a strongly orange-red compound - virtually an aniline dye. Cases have been described of orange-yellow coloration of urine following heavy exposure of workers to the dust of pendimethalin. Despite its structure as both a nitro-compound and aromatic amine, exposure to pendimethalin is NOT associated with methemoglobinemia.

Note: Have the product container or label with you when calling a poison control center or doctor or going for treatment.

5. Fire-Fighting Measures

Extinguishing media: Foam, carbon dioxide, dry chemical, and water spray.

Fires: Use as little water as possible. Dike area of fire to prevent pesticide run-off. Do not contaminate personnel or equipment, or handle broken packages or container without protective equipment. Avoid breathing vapors or fumes from burning materials, Control run-off water. If water enters drainage system advise authorities down stream.

Hazardous decomposition products:

Combustion may produce carbon and nitrogen oxides.

Protective clothing: Wear chemical goggles, rubber gloves and impermeable clothing. Wear a pesticide cartridge respirator.

6. Accidental Release Measures

Personal Precautions: Make sure all personnel involved in the spill cleanup follow good industrial hygiene practices. A small spill can be handled routinely. Wear suitable protective clothing and eye protection to prevent skin and eye contact. Use adequate ventilation and wear an

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air-supplied respirator to prevent inhalation.

Procedures for dealing with release or spill: Warning - flammable vapors may be present. Eliminate sources of ignition and ventilate spill area. Use non-sparking clean-up equipment to prevent vapor ignition. Control the spill at its source.

Contain the spill to prevent from spreading or contaminating soil or from entering sewage and drainage systems or any body of water. Clean up spills immediately, observing precautions outlined in Sections 7 and 8. Pump or scoop large amounts of liquid into a disposable container. Absorb remaining liquid or smaller spills with clay, sand or vermiculite. Scoop or sweep up material and place into a disposal container. Wash area with detergent and water. Pick up wash liquid with additional absorbent and place into compatible disposal container. On soils, skim off the upper contaminated layer and collect for disposal. Once all material is cleaned up and placed in a disposal container, seal container and arrange for disposition. Spillages or uncontrolled discharges into watercourses must be alerted to the appropriate regulatory body.

7. Handling and Storage

Handling practices: KEEP OUT OF REACH OF CHILDREN and animals. Prevent eating, drinking, tobacco use, and cosmetic application in areas where there is a potential for exposure to the material. Avoid breathing vapors or spray mist. Wear full protective clothing and equipment (see Section 8). After work, rinse gloves and remove protective equipment, and wash hands thoroughly with soap and water after handling, and before eating, tobacco use, drinking, or using the toilet. Wash contaminated clothing before re-use and separate from household laundry. Keep containers closed when not in use. Keep product, wash or rinse water, and contaminated materials out of water, away from crops, and away from access by people, animals and birds.

Appropriate storage practices/requirements: Store in original container only in a well-ventilated, cool, dry, secure area. Protect from heat, sparks and flame. Do not expose sealed containers to temperatures above 40°C. Keep separate from other products to prevent cross contamination. Rotate stock. Clean up spilled material immediately. National Fire Code classification: Not required

8. Exposure Controls/Personal Protection

Personal protective equipment for each exposure route:

General: Avoid breathing dust, vapors or aerosols. Avoid contact with eye, skin and clothing.

Wash thoroughly after handling and before eating, drinking, or handling tobacco.

INGESTION: Do not eat, drink, handle tobacco, or apply cosmetics in areas where there is a

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potential for exposure to this material. Always wash thoroughly after handling.

EYES:Where eye contact is likely, use chemical splash goggles. Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower.

SKIN: Where contact is likely, wear chemical-resistant gloves (such as nitrile or butyl), coveralls, socks and chemical-resistant footwear. For overhead exposure, wear chemical-resistant headgear.

INHALATION:A respirator is not normally required when handling this substance. Use process enclosures, local exhaust ventilation, or other engineering controls to keep airborne levels below exposure limits. A NIOSH-certified combination air-purifying respirator with an N, P or R 95 or HE class filter and an organic vapor cartridge may be permissible under certain circumstances where airborne concentrations are expected to exceed exposure limits. Protection provided by air-purifying respirators is limited. Use a pressure demand atmosphere-supplying respirator if there is any potential for uncontrolled release, exposure levels are not known, or under any other circumstances where air-purifying respirators may not provide adequate protection.

9. Physical and Chemical Properties

Appearance: Orange yellow liquid

Odor: Hydrocarbon odor

PH: 5.0 to 8.0

Flash point: More than 65°C

Vapor pressure and reference temperature:

1.94 mPa(25 °C) (Pendimethalin Technical).

Vapor density: Not available.

Boiling point: 183-210°C.

Melting point: Some crystallisation occurs between 0 and -7°C

Specific gravity or density: 0.97 @ 25°C

Evaporation Rate: Not available.

Water/oil partition coefficient: Not available.

Odour threshold: Not available.

Solubility in Water: Emulsifiable

Corrosiveness: Not Corrosiveness

10. Stability and Reactivity

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Chemical stability: Very stable in storage; store above 5 °C and below 130 °C. Stable to acids and alkalis. Slowly decomposed by light. DT₅₀ in water <21 d.

Conditions to avoid: Unstable under highly alkaline conditions. High temperatures, sparks, open flames. Keep away from sources of ignition.

Incompatibility with other materials: Strong oxidizing agents. Unstable under highly alkaline conditions.

Hazardous decomposition products: Mists from heated solution may cause respiratory irritation. Thermal decomposition products are toxic and may include hydrocarbons, oxides of carbon, nitrogen and fluorine compounds.

Hazardous polymerization: Will not occur.

11. Toxicological Information

Data for the formulated product:

Rat, Oral LD₅₀ (combined sexes) > 5000 mg/kg

Rabbit, Dermal LD₅₀ (combined sexes) > 2000 mg/kg

Rat, Inhalation LC₅₀ (4 hr): Not available

Rat, Inhalation LC₅₀ (1 hr calculated): Not available

Rabbit, Eye Irritation: Moderately irritating

Rabbit, Skin Irritation: Moderately irritating

Guinea pig, Dermal Sensitizer: Not a skin sensitizer.

Direct contact of this product with skin or sclera of eye may cause orange-yellow staining which will fade with time.

For pendimethalin: A marked depression in body weight gain and statistically significant increase in benign thyroid proliferative lesions were observed at the highest dose tested (5000 ppm) in the lifetime (24 month) rat study.

OSHA, NTP, or IARC Carcinogen: The US EPA has classified pendimethalin an oncogenicity rating of "low c" (possible human carcinogen).

12. Ecological and Ecotoxicological Information

Pendimethalin Technical, is practically non-toxic to insects (bees), but is slightly toxic to birds and toxic to fish and aquatic invertebrates (water flea).

Eco-Acute Toxicity

Pendimethalin Technical:

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Bees LC₅₀/EC₅₀ (contact) LD₅₀ (topical) >80.96µg/bee.

Invertebrates (Water Flea) LC₅₀/EC₅₀ (48 hr) 0.23 ppm

Fish (Trout) LC₅₀/EC₅₀ (96 hr) 0.11 ppm

Fish (Bluegill) LC₅₀/EC₅₀ (96 hr) 0.16 ppm

Birds (8-day dietary - Bobwhite Quail) LC₅₀/EC₅₀ >3,456ppm

Birds (8-day dietary - Mallard Duck) LC₅₀/EC₅₀ 1,369 ppm

Eco-Chronic Toxicity Pendimethalin Technical: Not Available

Environmental Fate The active ingredient, Pendimethalin, is relatively non-persistence in soil; reported half-life < 1 week. Pendimethalin breaks down or hydrolyzes rapidly in soils or water to the fluazifop acid, which is also of low persistence. Pendimethalin and fluazifop-p have low mobility in soils and not to present appreciable risks for groundwater contamination.

13. Disposal Considerations

Waste disposal information: Do not reuse empty containers. Empty container retains product residue. Triple rinse, or equivalent, empty container, return rinse water to dilution mixture, and dispose of dilution mixture as a hazardous waste if it cannot be disposed of by use according to label instructions. Dispose of empty containers in accordance with local regulations. Consult provincial environment ministry for advice on waste disposal. Industrial/commercial waste may be handled at licensed facilities only. Waste shipments must be securely packaged and properly labelled. Only licensed carriers may be used, and proper documents must accompany the shipment.

14. Transport Information

Shipping information such as shipping classification:

TRANSPORTATION OF DANGEROUS GOODS CLASSIFICATION - ROAD/RAIL Not Regulated.

Proper Shipping Name: Solid, N.O.S.

Hazard Class or Division: Class4.1

UN No.: 3082

PACKAGE GROUP: III

Identification Number: Not Regulated

15. Regulatory Information

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Symbol: Xn (Harmful)

Risk phrases: R10, R22

R10 Flammable.

R22 Harmful if swallowed.

Safety phrases: S1/2

S2 Keep out of the reach of children.

National legislation:

In accordance with 91/155/EEC Directive and with French standard T 01-102 and the South African Occupational Health and Safety Act, 1993 (Act. No. 85 of 1993).

16. Other Information

All information and instructions provided in this Material Safety Data Sheet (MSDS) are based on the current state of scientific and technical knowledge at the date indicated on the present MSDS and are presented in good faith and believed to be correct. This information applies to the product as such. In case of new formulations or mixes, it is necessary to ascertain that a new danger will not appear. It is the responsibility of persons on receipt of this MSDS to ensure that the information contained herein is properly read and understood by all people who may use, handle, dispose or in any way come in contact with the product. If the recipients subsequently produce formulations containing this product, it is the recipient's sole responsibility to ensure the transfer of all relevant information from this MSDS to their own MSDS.

